

Member ID: _____

Time: _____

Rank: _____



COMPUTER SECURITY

(320)

STATE 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each) _____ (100 points)

TOTAL POINTS _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

MULTIPLE CHOICE

Identify the letter of the choice that *best* completes the statement or answers the question.

1. How would you check the status of Linux firewall?
 - a. `sudo ufw status`
 - b. `iptables -L`
 - c. `systemctl is-enabled firewalld`
 - d. All of these are correct
2. If asked how to enable the firewall in Ubuntu 18.04 what command would you use?
 - a. `sudo ufw enable`
 - b. `systemctl start firewalld`
 - c. `sudo systemctl status firewalld`
 - d. `ufw enable`
3. What does ufw stand for?
 - a. Under Full Working (control)
 - b. Uncomplicated for Wifi (security)
 - c. Uncomplicated Fire Wall
 - d. Ubuntu Fire Wall
4. You need to deploy a VPN where you can whitelist the client MAC addresses to remain connected. Which do you use?
 - a. PPTP VPN
 - b. L2TP VPN
 - c. SSL VPN
 - d. IKEv2 VPN
5. Which of the following will help your customer decrease successful phishing attacks?
 - a. End user Training
 - b. YouTube Videos
 - c. Azure Multi-Factor Auth
 - d. Application Layer Gateway
6. You're consulting for a customer who reports users getting buffer overruns across the organization. How does Windows Defender protect against an exploitation like this?
 - a. Edge Browser
 - b. Data Execution Prevention (DEP)
 - c. Local security Policy
 - d. Defender Firewall

7. For security reasons, why would you use a service account?
 - a. So only the application can login to windows
 - b. So, a windows service will be registered, we use service accounts
 - c. A service account is used because it can start a service, but not login to windows
 - d. A service account is the same as a user account
8. Members of your Newspaper Reporting team (reporters) are multi-nationals, who surf the internet in at least 80 countries. What corporate policy should be employed, to mitigate security risk?
 - a. SOP
 - b. Role Based Training
 - c. Acceptable Use Policy
 - d. Personnel management
9. You find some competing reporters standing behind you, looking at your computer screen. What security step would be effective?
 - a. Management Meetings
 - b. Screen Filters
 - c. Human Resources complaint
 - d. Turn around and confront the problem
10. You have been asked to put a price on a recent malware attack on the corporation. What type of analysis is this calling for?
 - a. Quantitative
 - b. Qualitative
 - c. Measured Response
 - d. Accounting
11. You work in the espionage business and you know there is a certain number of exploits, which will result in loss every year. Describe the loss. In terms of risk management, which part of the equation is this?
 - a. Annualized loss expectancy (ALE)
 - b. Single Loss expectancy (SLE)
 - c. Annualized rate of occurrence (ARO)
 - d. Exposure Factor (EF)
12. You are conducting a report and find out the current incident was caused by an open proxy inside the office. What type of incident is this?
 - a. Accident
 - b. Flawed deployment
 - c. Resource Misconfiguration
 - d. Compromised resources

13. The incident response team member in charge of throwing out false positives and identifying actual threat is called the _____.
a. Specialist
b. Triage Analyst
c. Threat auditor
d. Administrator
14. You deploy the Cyber Incident Response team and they notify you the issue has grown over their ability to handle the problem. What must they do next?
a. Escalate the issue, according to the policies for such a contingency.
b. They must work around the clock until the issue is mitigated.
c. Collect information and analyze again.
d. Call the National Cyber Security Team in Washington DC.
15. When faced with an incoming packet, what is the first header the firewall should look at first?
a. Destination address
b. Mac address
c. Source Address
d. The GPS coordinates
16. When faced with an incoming packet, what is the first header the switch encounters?
a. Data Link Layer Ethernet Header
b. Ethernet Frame
c. Source IP address
d. Source Mac address and TCP address
17. BYOD (bring your own data) is a growing trend in business today. Which of the following best balance today's culture against the need for corporate protection?
a. Remote Wipe
b. Mobile Device Management MDM
c. Full Disk Encryption
d. Containerization
18. You suspect a company resource has been emailing secrets to a competitor. You need to retain an exact copy of the employee's hard drive. What native tools do you use?
a. Copy or Linux DD
b. Xcopy or DD
c. Ghost or Macrium
d. Third party software

19. You need to disable the ability of apps to know the location of a Windows 10 computer. Which is the correct GPO location?
- a. Computer Configuration > Windows Components > Data Collection and Preview Builds > Allow Telemetry
 - b. Computer Configuration > Windows Components > App Privacy > Let Windows apps access location
 - c. Computer Configuration > Windows Components > Find My Device > Turn On/Off Find My Device
 - d. Computer Configuration > Control Panel > Regional and Language Options > Allow users to enable online speech recognition services
20. Public DNS records that can mitigate the infiltration of phishing attacks in email include _____.
- a. MX and SPF Records
 - b. AAAA and MX Records
 - c. DNSSEC and DMARC Records
 - d. Reverse DNS and A records
21. Which of the NIST incidence response steps is considered reactive?
- a. Preparation
 - b. Detection and Analysis
 - c. Containment, Eradication, and Recovery
 - d. Post-Incident Activity
22. There are several types of authentication attacks: brute force, credentials stuffing, password spraying and others. Which definition below matches the description for password spraying?
- a. Use of a single user account and a methodology is used to try a significant number of passwords.
 - b. Use of several user accounts, with almost random attempts to login to each, with varied frequency. This is generally paired with lists of common passwords.
 - c. Use of a known password to attempt to access areas to gain additional passwords or increased level of access from the already stolen credentials.
 - d. Use of 5-10 User accounts with login attempts tried every 45 seconds. Upon failure, another user/password attempt is tried. This goes on for 48 minutes or until the user list is completed.
23. On a stand-alone Windows 10 machine, which cloud service would you most likely experience as a personal user?
- a. SAAS Software as a Service
 - b. IAAS Infrastructure as a Service
 - c. PAAS Platform as a Service
 - d. DAAS Data as a Service

24. This type of malware does not rely on a virus laden file structure on a hard disk. What kind of malware is this, and where does it reside?
- Trojan Malware- Registry
 - Worm Malware - Bios
 - Virus Malware- Internet
 - File less Malware- Memory
25. You need to filter out viruses and malware from Internet connections in the office. What is the best means to do so?
- DMZ
 - Proxy Server
 - Packet Switched Network
 - Enable Screening
26. What new security item is part of Windows after version 1703?
- Windows Defender
 - Windows Defender Security Center
 - Windows Firewall
 - Windows PowerShell security Snap in
27. How can you track down a virus running from the browser and potentially stop it?
- Task Manager, Processes, Open containing folder, Search registry, Stop service, Delete files and registry
 - Format the drive and clean install.
 - cmd, netsh winsock reset, devmgmt.msc and delete any bangs
 - Boot outside the OS and hit F8 to boot to Service Restore mode.
28. One kind of key is used for 1x1 encryption and the other is used for 1 to many encryption. Name both of these methods.
- public/Private
 - SSL/TLS
 - symmetric/Asymptotic
 - Symmetric/Asymmetric
29. What is a windows/Azure Security Principal?
- A User, Computer or Service
 - The Token given to a Logged in user
 - The SPN of a computer
 - A users GUID
30. What is the difference between secpol.msc and gpedit.msc?
- They are the same thing.
 - secpol is a subset of gpedit
 - gpedit is a subset of secpol.
 - They are not related

31. You can enable usb write protection, for usb drives, by going to the registry and adding a DWORD called StorageDevicePolicies in computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\, and then adding a WriteProtect key with a value of what?
- 1
 - 0
 - 3
 - 4
32. What is mimikatz used for?
- It is a common hack tool for saving cached passwords and hashes
 - It is an internet download for Azure
 - It is an Exploit tool used with viruses
 - It can be used to test hardening of your environment
33. Is Internet Explorer 11 Still supported on windows 10?
- Yes, but it ends June 15th 2022.
 - No, Chrome has replaced it.
 - No, Microsoft uses Edge now.
 - Yes, it will support IE11 into the future.
34. What is this called in the Windows nomenclature \\localhost\c\$
- Administrative Share
 - The Back Door Share
 - The Hackers Way
 - The Windows Share
35. What is the *best* practice when working with accounts of terminated users?
- delete the account
 - change the password
 - do nothing
 - disable the account
36. Which computer security word *best* describes checking your driver's license to verify you are who you say?
- Authenticated
 - Authorized
 - Login Credential
 - Password
37. What program should be used to manage firewall rules in Windows?
- Malware Bytes Anti-Malware
 - Windows Firewall
 - Security Center
 - Windows Defender Firewall with Advanced Security

38. If ping fails to your machine, you need to create a rule in windows firewall. Which one?
- Allow - (Echo Request – ICMPv4-In)
 - Enable File and Printer Sharing (Echo Reply ICMPv4-in)
 - Core Networking – IPv6 (IPv6-In)
 - File and Printer Sharing (LLMNR-UDP-In)
39. What technology built-in to Windows allows users to encrypt their files and volumes?
- RSA
 - File Explorer
 - CryptKey
 - Bitlocker
40. Which of the following is *not* a characteristic of a complex password?
- 5 or more characters
 - lowercase
 - numbers
 - special characters
41. What technology is used to scan a fingerprint for authentication?
- three-factor authentication
 - scanner
 - biometrics
 - smart card
42. What character can be used to create a hidden file share?
- @
 - \$
 - %
 - *
43. What type of electronic document contains a public key?
- digital certificate
 - biometrics
 - PIN
 - PAN
44. What do you call multiple Windows updates that have been packaged together as one installation and are well tested?
- service packs
 - cumulative packs
 - critical update
 - optional update

45. At what layer of the OSI model does the IP protocol function?
- a. Transport Layer
 - b. Network Layer
 - c. Data Link Layer
 - d. Presentation Layer
46. What protocol can be used by a host on a network to find the MAC address of another device based on an IP address?
- a. DNS
 - b. ARP
 - c. TCP
 - d. UDP
47. What type of device looks at packets and forwards it based on its destination IP address?
- a. bridge
 - b. switch
 - c. router
 - d. hub
48. What item, about the size of a credit card, allows access to a network and its resources?
- a. digital certificate
 - b. smart card
 - c. security token
 - d. PIN
49. What term describes taking advantage of someone's natural tendencies and emotional responses in order to gain access to data?
- a. phreaking
 - b. hacking
 - c. social engineering
 - d. reverse engineering
50. What type of malware can copy itself and infect a computer on every reboot, without the user's consent or knowledge?
- a. virus
 - b. Trojan horse
 - c. rootkit
 - d. backdoor